



Minimum Load Shedding



Scope. A deeper look at the *minimum load shedding* (MLS) problem — the optimization that defines the consequence function $C(s)$ in probabilistic risk assessment. We cover the DC and AC formulations, objective variants, convexity and duality, solution methods, and the practical subtleties that change risk estimates.

1. Why minimum load shedding sits at the heart of PRA

In a PRA loop you sample a system state s (a contingency, a load/renewable realization) and ask: *given this state, what is the unavoidable consequence?* The honest answer is **the least amount of load that must be curtailed so that the remaining network operates within its physical and security limits**. That minimum is the consequence $C(s)$, and aggregating it over states yields EENS, LOLP/LOLE, and tail metrics.

Two framings coexist:

- **Feasibility test.** Is there *any* operating point that serves all load within limits? If yes, $C(s) = 0$ (the state is healthy). If no, the state is a *failure* and we must shed.
- **Optimal recourse.** When full service is infeasible, find the redispatch + curtailment that restores feasibility while shedding as little (or as little *value*) as possible.

Minimum load shedding answers both at once: the optimal objective is zero exactly when the state is feasible, and positive otherwise.

2. The DC formulation (the workhorse LP)

Let buses be indexed by $i \in \mathcal{N}$, lines by $\ell \in \mathcal{L}$, generators by $g \in \mathcal{G}$. With the linearized (DC) network, the canonical minimum-load-shedding LP is:

$$\min_{p, \theta, \Delta d} \sum_{i \in \mathcal{N}} \Delta d_i$$

subject to

$$\sum_{g \in \mathcal{G}_i} p_g - (d_i - \Delta d_i) = \sum_{\ell \in \delta(i)} f_\ell, \quad \forall i \in \mathcal{N}$$

$$f_\ell = \frac{1}{x_\ell} (\theta_{i(\ell)} - \theta_{j(\ell)}), \quad |f_\ell| \leq f_\ell^{\max}, \quad \forall \ell \in \mathcal{L}$$

$$0 \leq p_g \leq \bar{p}_g \quad \forall g, \quad 0 \leq \Delta d_i \leq d_i \quad \forall i, \quad \theta_{\text{ref}} = 0$$

where d_i is nominal demand at bus i , Δd_i the load curtailed there, p_g generator output, θ_i voltage angle, f_ℓ line flow, x_ℓ reactance, and $f_\ell^{\max}$ the thermal limit. The objective $\sum_i \Delta d_i$ is total MW shed.



Why this is attractive. It is a **linear program**: convex, always feasible (you can always shed *all* load, $\Delta d_i = d_i$), and fast enough to solve millions of times inside Monte Carlo. The optimum is unique in objective value, giving a well-defined $C(s)$.

Including ramping / reserves

When the pre-contingency dispatch p^0 matters, generator deviations are bounded by ramp/reserve limits: $|p_g - p_g^0| \leq R_g$. This couples the post-contingency recourse to the base case and is essential for *operational* (as opposed to adequacy) risk.

3. Objective variants

The choice of objective is a modeling decision that materially changes which states look "risky."

Objective	Expression	Interpretation
Minimum MW shed	$\min \sum_i \Delta d_i$	Treats every MW as equal; yields EENS directly
Minimum cost / VOLL-weighted	$\min \sum_i v_i \Delta d_i$	v_i = value of lost load at bus i ; prioritizes critical load
Priority / lexicographic	shed tier-3 before tier-2 ...	Enforces firm/interruptible contract order
Min generation + shedding cost	$\min \sum_g c_g p_g + \sum_i v_i \Delta d_i$	Full security-constrained OPF with curtailment as last resort
Min number of buses shed	$\min \sum_i 1[\Delta d_i > 0]$	Fairness/operational simplicity; turns the LP into a MILP

With a pure $\min \sum v_i \Delta d_i$ objective and high enough v_i relative to generation cost, the solver will exhaust all redispatch before shedding any load — so MLS becomes the *curtailment tail* of a security-constrained OPF.

4. The AC formulation

The DC model ignores voltages, reactive power and losses — exactly the physics behind voltage-collapse-driven blackouts. The AC minimum-load-shedding problem restores them:

$$\min \sum_i v_i \Delta d_i$$

subject to the AC power-flow balance at every bus

$$P_i(V, \theta) = \sum_{g \in \mathcal{G}_i} p_g - (d_i - \Delta d_i), \quad Q_i(V, \theta) = \sum_{g \in \mathcal{G}_i} q_g - (q_i^d - \Delta q_i)$$

with

$$P_i = V_i \sum_k V_k (G_{ik} \cos \theta_{ik} + B_{ik} \sin \theta_{ik}), \quad Q_i = V_i \sum_k V_k (G_{ik} \sin \theta_{ik} - B_{ik} \cos \theta_{ik})$$

and operating limits

$$\underline{V}_i \leq V_i \leq \overline{V}_i, \quad \underline{q}_g \leq q_g \leq \overline{q}_g, \quad |S_\ell| \leq S_\ell^{\max}.$$

Here V_i is voltage magnitude, $\theta_{ik} = \theta_i - \theta_k$, and G_{ik}, B_{ik} are the real/imaginary parts of the bus admittance matrix. To keep load power factor realistic, real and reactive curtailment are often linked: $\Delta q_i = \Delta d_i \tan \phi_i$.



The catch: nonconvexity. The AC balance equations are nonconvex, so the MLS problem is an NLP that can have local optima and — importantly — a *genuinely infeasible full-service point* is the signal that load must be shed.

Convex relaxations

To recover tractability and global bounds, the AC feasible set is commonly relaxed:

- **SOCP relaxation** (second-order cone) — tight on radial/distribution networks; fast.
- **SDP relaxation** (semidefinite) — often tight on meshed transmission; heavier.
- **QC and convex-DistFlow** relaxations — intermediate options.

If the relaxation is exact, the recovered point is globally optimal; if not, its objective is a valid lower bound on the true minimum load shed, which is useful for *bounding* risk.

5. Structure, duality, and what the solution tells you

For the DC LP, the dual carries operational meaning:

- The **dual of the nodal balance** is the locational marginal price of energy at each bus; in a shedding regime its magnitude relates to the value of lost load and reveals which buses are *binding*.
- The **duals of the line limits** are congestion prices; nonzero entries identify the constraints actually responsible for the curtailment.
- **Complementary slackness** tells you that load is shed only at buses that cannot be reached by cheaper recourse — a compact explanation of *why* a state fails.

This is why MLS doubles as a diagnostic: the active constraint set and shadow prices localize the weakness (a thermal bottleneck, a reactive-power deficiency, an islanded pocket).

6. Islanding, topology, and discrete recourse

Real contingencies often **split the network into islands**. Each island must balance generation and load independently, frequently triggering shedding in import-dependent islands even when system-wide capacity is adequate. Extensions that change the optimum:

- **Transmission switching** — opening lines as a remedial action (adds binaries → MILP).
- **Under-frequency / under-voltage load shedding (UFLS/UVLS)** — the automatic, block-wise relay schemes the real system uses; discrete and location-fixed, unlike the idealized continuous LP.
- **Discrete load blocks** — curtailment in fixed increments rather than continuous Δd_i , again turning the LP into a MILP.



Idealization gap. The continuous LP/NLP gives the *theoretical minimum* shed and thus a (often optimistic) lower bound on consequence. Real relay-based schemes shed in coarse blocks and can over-shed. PRA studies should be explicit about which they model, since it biases EENS.

7. Computation inside the PRA loop

MLS is solved once per sampled state, so performance dominates study feasibility:

- **Warm-starting** from the base case or neighboring contingencies (the simplex/interior-point reuses prior bases).
- **Parametric / sensitivity** updates: many contingencies are rank-one changes to the base LP.

- **Screening** cheap DC-MLS first, escalating to AC only for states near the feasibility boundary.
- **Decomposition** (Benders) when MLS is the recourse stage of a two-stage stochastic SCOPF/unit-commitment problem.
- **ML surrogates / active learning** to predict feasibility or the shed amount and skip easy states.

8. A compact worked skeleton (DC)

For a single contingency, the smallest useful instance is just:

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variables:  p_g  (gen output),  d_shed_i (>=0, <= d_i),  theta_i
minimize:   sum_i  v_i * d_shed_i
subject to: nodal balance (PTDF or B-theta form)
            |f_l| <= f_l_max    for all lines
            0 <= p_g <= pbar_g  for all gens
            |p_g - p0_g| <= R_g  (optional ramp)
result:     objective = 0  -> state feasible (C(s)=0)
            objective > 0  -> minimum load shed = C(s)
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Using a **PTDF (shift-factor) form** eliminates the angles θ and expresses flows directly as $f = \text{PTDF} \cdot (\text{injections})$, which is convenient for fast repeated solves and for embedding line limits as a compact constraint matrix.

9. Pitfalls that bias risk estimates

- **DC vs. AC mismatch** — DC-MLS can declare a state feasible that AC voltage limits would force to shed; it systematically *under*-estimates EENS in reactive-constrained systems.
- **Ignoring islanding** — system-wide balance hides local shortfalls.
- **Continuous vs. block shedding** — idealized continuous curtailment under-states real over-shedding.
- **Objective choice** — minimizing MW vs. minimizing value can rank contingencies differently; pick the one matching the decision.
- **Relaxation inexactness** — a non-exact SOCP/SDP solution is a bound, not the operating point; don't report it as the achieved shed.
- **Cascading neglect** — MLS gives the *one-shot* optimal recourse; it does not by itself capture subsequent overload-driven trips unless embedded in a cascade loop.